



INVENTORY MANAGEMENT SYSTEM USING SAP ABAP WITH ALV

AN INTERACTIVE AND INTELLIGENT INVENTORY MONITORING SOLUTION



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ABSTRACT

This project presents a backend-based Inventory Management System developed using SAP ABAP. The system integrates data from multiple SAP tables such as MARD, MARA and MAKT to provide real-time inventory insights. Data is processed efficiently using internal tables and displayed using an interactive ALV (ABAP List Viewer) grid. Key features include traffic-light indicators for stock levels, KPI-based analysis and drill-down functionality for material movement tracking. The system transforms raw SAP data into structured, interactive reports, enabling faster and more effective decision-making in inventory management.

INTRODUCTION

Inventory management is essential for maintaining balance between supply and demand in organizations. SAP ERP systems store large volumes of inventory data, but extracting insights requires efficient backend processing and reporting tools.

This project develops a smart inventory management system using SAP ABAP to:

- Integrate data from multiple SAP tables
- Process inventory data efficiently
- Provide structured and interactive reports
- Support real-time decision-making

The system enhances data visibility by transforming raw inventory data into meaningful insights. It reduces manual effort and improves accuracy in stock analysis.

RESULTS AND ANALYSIS

Inventory Management

P_WERKS
P_MATNR
P_LGORT

Figure 1: Selection screen for entering inventory parameters such as Plant, Material, and Storage Location.

Inventory Management

Material	Description	Material Type	Plant	Storage Location	Stock
OC T-FL131	Drive Assembly	HALB	1010	101A	10,000
OC T-FL131	Antrieb	HALB	1010	101A	10,000
OC T-FL131	Transmission	HALB	1010	101A	10,000
OC T-FL132	Drive Assembly	HALB	1010	101A	10,000
OC T-FL132	Antrieb	HALB	1010	101A	10,000
OC T-FL132	Transmission	HALB	1010	101A	10,000
OC T-FL133	Drive Assembly	HALB	1010	101A	10,000
OC T-FL133	Antrieb	HALB	1010	101A	10,000
OC T-FL133	Transmission	HALB	1010	101A	10,000
OC T-FL134	Drive Assembly	HALB	1010	101A	10,000
OC T-FL134	Antrieb	HALB	1010	101A	10,000
OC T-FL134	Transmission	HALB	1010	101A	10,000
OC T-FL135	Drive Assembly	HALB	1010	101A	10,000
OC T-FL135	Antrieb	HALB	1010	101A	10,000
OC T-FL135	Transmission	HALB	1010	101A	10,000
OC T-FL136	Drive Assembly	HALB	1010	101A	10,000
OC T-FL136	Antrieb	HALB	1010	101A	10,000
OC T-FL136	Transmission	HALB	1010	101A	10,000
OC T-FL137	Drive Assembly	HALB	1010	101A	10,000
OC T-FL137	Antrieb	HALB	1010	101A	10,000
OC T-FL137	Transmission	HALB	1010	101A	10,000

Figure 2: Interactive ALV report displaying structured inventory data with stock levels and material details.

Inventory Management

*** INVENTORY SUMMARY ***
Total Materials: 500
Low Stock Items: 500

Figure 3: Summary of key metrics including total materials and low stock items for quick analysis.

Inventory Management

Material	Document	Quantity	Move Type	Move Description	Posting Date
T-FL131	4900001476	10,000	561	SC vendor consignment	16.08.2023
T-FL131	4900001476	10,000	561	SC RTP	16.08.2023
T-FL131	4900001476	10,000	561	SC project stock	16.08.2023
T-FL131	4900001476	10,000	561	SC project stock	16.08.2023
T-FL131	4900001476	10,000	561	SC sales order	16.08.2023
T-FL131	4900001476	10,000	561	SC sales order	16.08.2023
T-FL131	4900001476	10,000	561	In.ent.st.bals:proj.	16.08.2023
T-FL131	4900001476	10,000	561	In.ent.st.bals:slsOrd	16.08.2023
T-FL131	4900001476	10,000	561	SC customer stock	16.08.2023
T-FL131	4900001476	10,000	561	In.ent.st.bals:cst	16.08.2023
T-FL131	4900001476	10,000	561	In.ent.st.bals:cuCons	16.08.2023
T-FL131	4900001476	10,000	561	In.ent.st.bals:cuRePa	16.08.2023
T-FL131	4900001476	10,000	561	In.ent.st.bals:SC stk	16.08.2023
T-FL131	4900001476	10,000	561	In.ent.st.bals:RIP	16.08.2023
T-FL131	4900001476	10,000	561	In.ent.st.bals:consgrt	16.08.2023
T-FL131	4900001476	10,000	561	Init.entry of stBal.	16.08.2023
T-FL131	4900001476	10,000	561	In.ent.st.bals:slsOrd	16.08.2023
T-FL131	4900001476	10,000	561	SC vendor consignment	16.08.2023
T-FL131	4900001476	10,000	561	SC RTP	16.08.2023
T-FL131	4900001476	10,000	561	SC project stock	16.08.2023

Figure 4: Detailed view of material movement history including document number, quantity, and posting date.

PROBLEM STATEMENT

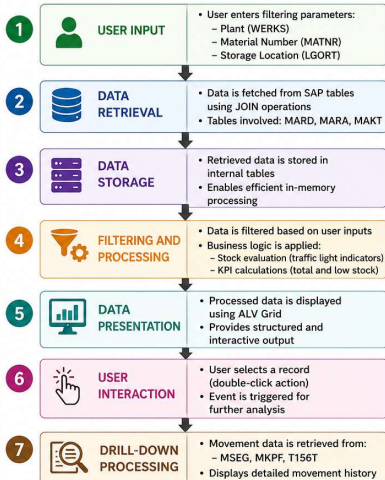
- Inventory data in SAP systems is distributed across multiple tables, making it difficult to access and analyze efficiently.
- Traditional reporting methods lack real-time visibility and do not provide interactive analysis capabilities.
- Manual data handling leads to errors, inefficiency, and delays in decision-making.
- There is a need for a system that can integrate, process, and present inventory data in a structured, interactive, and user-friendly manner.

KEY FEATURES

- Real-time inventory monitoring using SAP data
- Integration of multiple SAP standard tables (MARD, MARA, MAKT, MSEG)
- Interactive ALV (ABAP List Viewer) reports with sorting and filtering
- Traffic-light indicators for stock status identification & KPI-based analysis for better decision-making
- Drill-down functionality for detailed material movement tracking & Efficient data processing using internal tables
- Dynamic user-based filtering (Plant, Material, Storage Location)

METHODOLOGY

The system processes data through the following structured sequence:



SYSTEM ARCHITECTURE

PRESENTATION LAYER

- User interacts through selection screen
- Input Parameters: WERKS, MATNR, LGORT
- Output displayed using ALV grid
- Features: Sorting, Filtering, Interactive display

APPLICATION LAYER

- Core logic implemented using ABAP
- Data retrieval using SQL JOINS
- Data processing using internal tables
- Business Logic:
 - Stock evaluation (traffic lights)
 - KPI calculation
- Handles user actions (drill-down events)

DATA LAYER

- Stores and manages inventory data
- SAP Tables Used:

MARD	MARA	MAKT	MSEG	MKPF	T156T
Stock Data	Material Data	Descriptions	Movement Data	Header Data	Movement Types

CONCLUSION & FUTURE SCOPE

Conclusion:

The developed system provides an efficient and scalable solution for inventory monitoring using SAP ABAP. It integrates data from multiple SAP tables and transforms it into structured, meaningful insights through interactive ALV reports. The use of KPI analysis and traffic-light indicators enables quick identification of stock conditions, improving decision-making and reducing manual effort.

Future Scope:

- Development of a modern SAP Fiori-based UI
- Integration of graphical dashboards and data visualization
- Implementation of automated alerts for low stock conditions
- Integration with additional SAP modules (Sales, Procurement)
- Performance optimization using advanced techniques like CDS views
- Mobile accessibility for real-time inventory monitoring

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PROJECT
REPORT

